COMP3011 – WEB SERVICES AND WEB DATA

UK Transport Reliability API

Data-Driven Incident Management and Reliability Analytics

Feng He

Coursework 1 Oral Presentation

PROBLEM & MOTIVATION

Why this project?

Public transport disruptions significantly impact **journey reliability** and user experience. While raw transport feeds exist, they are often difficult to interpret or use directly for long-term planning.

Project Goal

Build a database-backed web API that moves beyond static data to support **real-time analytics** and proactive incident management.

WHAT THE API DOES

Core Operations

Full Incident CRUD operations paired with automated TfL Tube status ingestion to maintain a live system state.

Reliability Analytics

Advanced reliability scoring, delay heatmap generation, and anomaly detection for transport corridors.

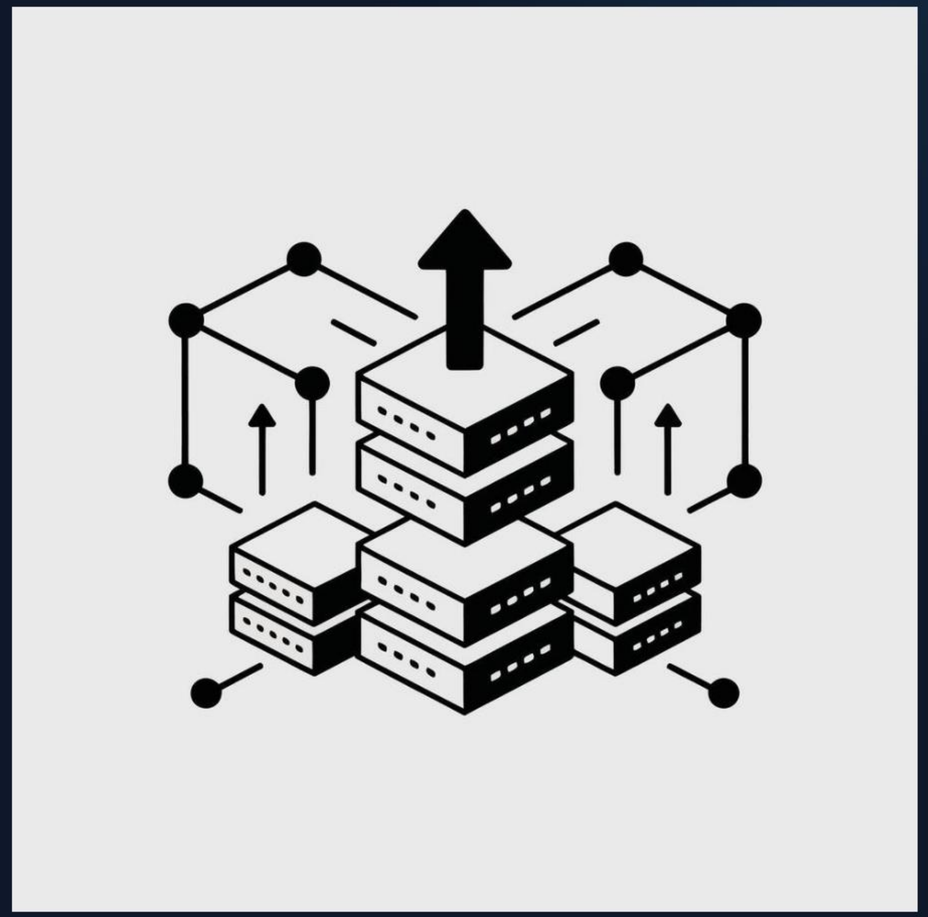
Agent-Ready

Goes beyond coursework basics by adding an analytics layer and an LLM-agent integration story.

ARCHITECTURE & STACK

The Tech Stack

FastAPI: High-performance async HTTP handling.
PostgreSQL: Robust relational storage for spatial analytics.
SQLAlchemy: Efficient ORM management.
Docker: Standardised environment execution.



Layered design ensures separation of concerns between logic and data persistence.

DATA MODEL & ENDPOINTS

Category	Endpoint / Entity Details	Purpose
Model	Incident: ID, Mode, Line, Station, Severity, Delay	Core storage entity
CRUD	POST /incidents | GET /incidents	Manage incident lifecycle
Query	GET /incidents/{id} | PATCH /incidents/{id}	Specific resource access
Clean	DELETE /incidents/{id}	Resource removal

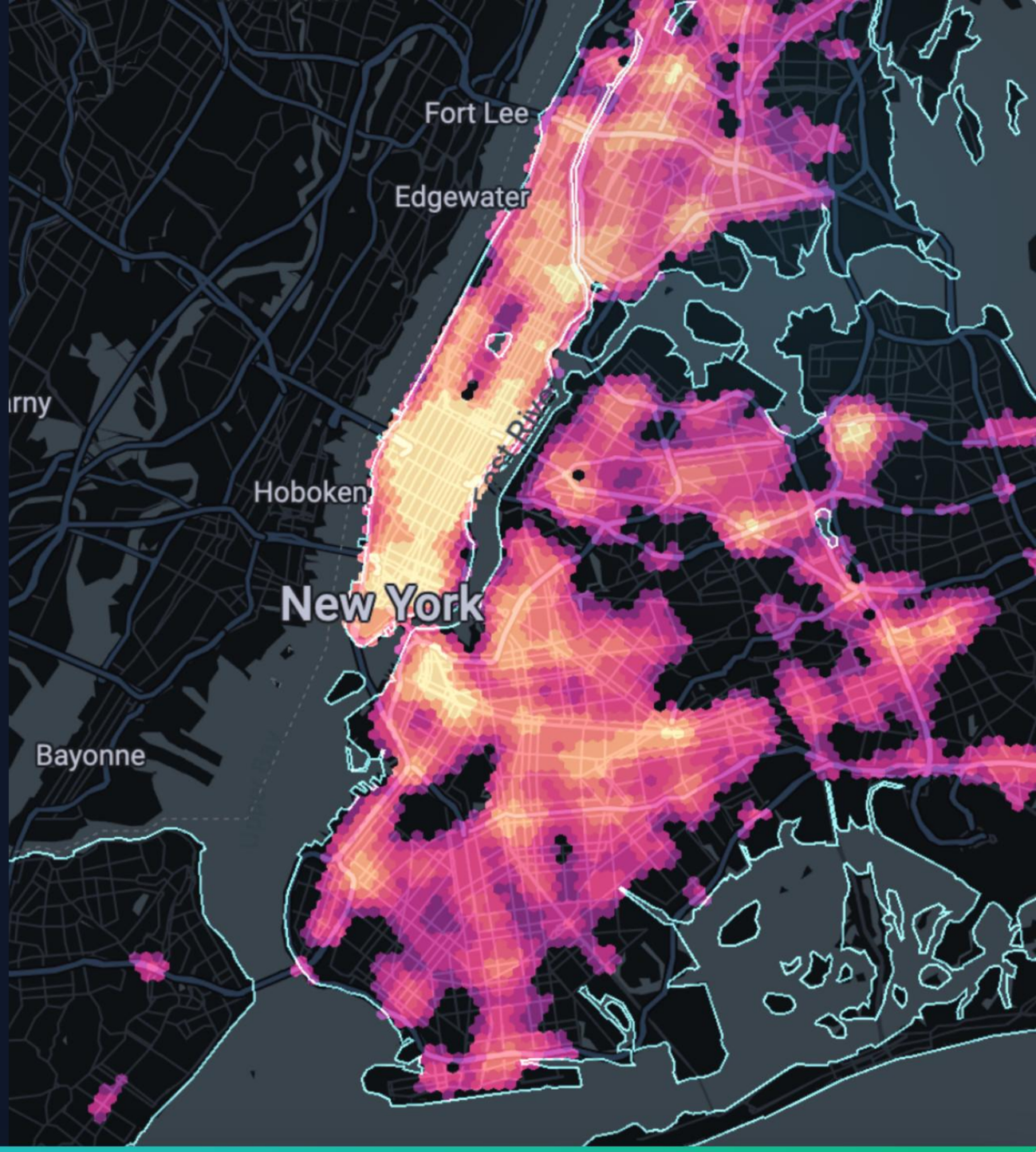
Fingerprinting is implemented to ensure automatic deduplication of upstream feeds.

SYNC & ANALYTICS

Turning Data into Insight

The API automates the **normalization of TfL data**, storing only significant disruptions.

- **Live Sync:** Ingests Tube status via /sync/tfl.
- **Heatmaps:** Identifies high-friction transit nodes.
- **Anomalies:** Detects statistically rare delays.



DEMONSTRATION FLOW

1. Seed

Bootstrap test data via /dev/seed

2. Ingest

Sync live TfL Tube statuses

3. Query

Filter active incidents by line

4. Analyze

Generate heatmap & reliability score

5. Detect

Identify anomalies in historical data

TESTING & ERROR HANDLING

201 Created

Successful incident creation and data seeding.

404 Not Found

Proper handling of invalid incident lookups.

502 Upstream

Handling failures in the external TfL API feeds.

CHALLENGES & FUTURE WORK

Challenges

- Balancing coursework compliance with ambitious analytics.
- Maintaining a clear, explainable design during data sync.

Future Road Map

- Implementing OAuth2 authentication & ownership.
- Expanding coverage to all UK rail operators.
- Adding predictive ML for incident duration forecasting.

Any Questions?

UK Transport Reliability API

Note on GenAI: Used for architectural planning and report structuring.
Final implementation & validation by the author.

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IMAGE SOURCES

https://media.istockphoto.com/id/1350389827/photo/an-underground-train-in-motion.jpg?s=612x612&w=0&k=20&c=E13COXpdeTwYmaSAm6SELtjVvweJRfNpaESwY-gMtPw=

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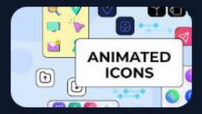

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